

**DRAFT**

Do Not Use Until Posted.

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**Question #: 1**

A researcher is studying genetic disorders caused by connective tissue defects and is focusing on the covalent cross-links that are found in collagen and elastin. These cross-links increase the stability and strength of connective tissue. The cross-linking reaction occurs when specific amino acid residues are post-translationally modified in these proteins. Which of the following amino acids most likely undergoes this process?

- A. Glycine
- ✓B. Lysine
- C. Proline
- D. Alanine
- E. Cysteine

**Rationale:** Lysine is oxidized by lysyl oxidase (copper containing) to form allysine residues which covalently link to form cross-links in elastin and collagen. Proline undergoes hydroxylation to form hydroxyproline residues (by prolyl hydroxylase requiring vitamin C) which are required for the formation of hydrogen bonds in collagen.

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**Question #: 2**

An investigator is studying components of connective tissue. A biopsy from a healthy volunteer shows the proteoglycan aggregates isolated from the extracellular matrix. Which of the following best describes the isolated macromolecules?

- A. Consist of approximately 90% protein
- B. Highly positively charged molecules that results in considerable electrorepulsion
- C. Contain long branched sugar units
- ✓D. Use linker proteins to form aggregates with hyaluronic acid as central strand
- E. Assembled to proteoglycan aggregates entirely in the endoplasmic reticulum

**Rationale:** Proteoglycans contain GAGs (about 90-95%). They are negatively charged macromolecules due to the presence of acidic sugars and sulfates. The Glycosaminoglacans are unbranched carbohydrates that contain a repeating disaccharide unit. Assembly of proteoglycan aggregates occurs in the extracellular matrix. The core proteins in proteoglycan monomers are connected to the central

hyaluronic acid strand in proteoglycan aggregates via linker proteins.

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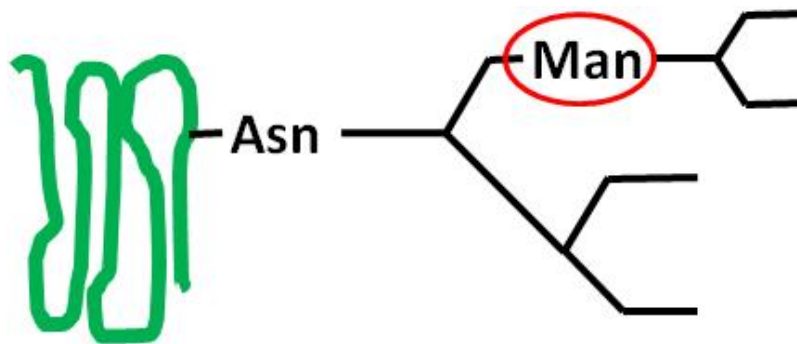
**Question #: 3**

A 2-month-old girl is brought to the physician by her parents because of a delay in developmental milestones. Laboratory studies show a defect in sorting the enzymes to the lysosomes. Additional studies show a defect in the modification of the lysosomal enzyme as shown in the figure. Which of the following modifications of the sugar is most likely required for appropriate sorting of this enzyme?

- A. Acetylation
- B. Farnesylation
- C. Oxidation
- ✓D. Phosphorylation
- E. Ubiquitination
- F. Hydroxylation

**Rationale:** The enzyme shown in the figure is an N-linked glycoprotein (carbohydrate linked to asparagine). Phosphorylation of mannose residues (mannose 6-Phosphate) in the Golgi results in the sorting of the enzymes to the lysosomes. A defect in this process results in I-cell disease

**Attachment:**



58628.JPG

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**Question #: 4**

A 20-year-old woman comes to the physician because of a 3-day history of bleeding gums while brushing her teeth. She has been on a special diet for the past three weeks with foods low in carbohydrates and deficient in fruits and vegetables. She does not take any supplements. Which of the following enzymes is most likely lacking its coenzyme and is responsible for her symptoms?

- A. Lysyl oxidase
- ✓B. Prolyl hydroxylase
- C.  $\gamma$ -glutamyl carboxylase
- D. Transketolase
- E. Glucose 6-phosphate dehydrogenase

**Rationale:** The vignette describes a patient with vitamin C deficiency. Vitamin C is rich in fresh fruits and vegetables. It is required as a coenzyme for prolyl hydroxylase. Addition of hydroxyl groups to collagen increases the formation of hydrogen bonds and hence increases collagen stability. Lysyl oxidase requires copper. Thiamine (vitamin B1) is a coenzyme for transketolase. Gamma glutamyl carboxylase requires vitamin K.

#### Question #: 5

A male newborn delivered at full-term is being evaluated for bilateral hip dislocation. Physical examination shows velvety skin and increased joint mobility. Laboratory studies show abnormal processing of collagen in the extracellular matrix. Which of the following processes is most likely affected in this patient?

- A. Hydroxylation of collagen
- B. Hydrogen bonding in the triple helix
- C. Glycosylation of collagen
- D. Formation of procollagen triple helix
- ✓E. Cleavage of procollagen

**Rationale:** The question is testing for the process in collagen synthesis which occurs in the extracellular matrix.

The various steps of collagen synthesis and their intracellular location is listed below:

Nucleus: transcription of collagen genes: for Type I collagen: COL1A1 and COL1A2 to synthesize mRNA

Rough Endoplasmic Reticulum:

1. Synthesis (translation of mRNA) and entry of pro- $\alpha$ -chains into the ER lumen.
2. Hydroxylation of specific proline and lysine residues (vitamin C).
3. Start of O-glycosylation of selective hydroxylysine residues with glucose or galactose.
4. Chaperones: Linkage of the 3 correct  $\alpha$ -chains at the C-terminal by disulfide bonds ( $\alpha_1(\text{I})$   $\alpha_2(\text{I})$ ) and formation of the procollagen triple-helix (C- to N-terminal zipper). Closing of triple-helix with disulfide bonds at the N- terminal propeptides.

Golgi: Finishing glycosylation; and procollagen is packed into secretory vesicles for release into the ECM.

ECM: Cleavage of propeptides by propeptidases leads to the insoluble collagen (tropocollagen). Staggered self-assembly of the collagen molecule. Covalent extracellular cross-linking using lysyl oxidase (copper) which forms allysine and hydroxyallysine residues in collagen.

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**Question #:** 6

An investigator uses human cell culture to study the role of enhancer proteins in transcriptional regulation. They note that several classes of trans-acting factor proteins coordinate to influence gene expression in this eukaryotic system. Which of the following statements best describes the target of enhancer proteins that function in this type of regulation?

- A. An mRNA sequence
- ✓B. A DNA sequence
- C. RNA polymerase II
- D. The ribosome
- E. The RISC complex
- F. The first stem-loop of an miRNA
- G. A tRNA
- H. An rRNA
- I. DNA gyrase
- J. DNA helicase
- K. DNA ligase

**Rationale:** Trans-acting factors are proteins which bind to specific DNA sequences to regulate gene expression. Enhancers are trans-acting factors (proteins) which bind to DNA to regulate gene expression. Importantly, the sequence that enhancers might bind to have the ability to function at great distances, sometime many kilo bases away from the gene in which they regulate. Enhancer sequences (the DNA) may be upstream, downstream, or within the actual gene that they regulate. The ability that allows this "regulation at a distance" is thought to be mediated by DNA looping, and chromatin domains.

RNA sequences are synthesized by RNA polymerase from DNA templates. The ribosome is a protein-RNA complex which translates mRNA into polypeptide. The RISC (RNA induced silencing complex) is involved in suppressing gene expression.

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**Question #:** 7

A researcher studying E. coli lactose metabolism finds that a main regulatory aspect involves a polycistronic mRNA. Gene expression studies show that this operon is induced only under the appropriate environmental and metabolic

conditions. Which of the following best explains the proper function of this gene regulatory system?

- A. The inducer of the *lac* operon is cAMP
- B. CAP/CRP protein only binds to the promoter region in the presence of glucose
- ✓C. The repressor protein LacI, only binds the operator in the absence of lactose
- D. A non-functional mutation of the operator will prevent gene expression
- E. The *lac* operon is only expressed in the presence of both lactose and glucose

**Rationale:** The repressor protein binds to the operator (a DNA sequence) in the absence of lactose until the repressor senses the presences of allolactose, which is an isomer of lactose.

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**Question #: 8**

A newly married couple comes to the physician for advice because the man has Phenylketonuria (PKU), which has been treated with life-long dietary restriction. The woman is healthy, and their relationship is not consanguineous. In terms of the standard nomenclature for human recessive and dominant disorders, which of the following will be the most likely genotype of their children?

- A. AA
- ✓B. Aa
- C. aa
- D. 50% of their children will be AA and 50% will be Aa
- E. May not be predicted from the information in the question
- F. Disorder is multifactorial, so genotype information is irrelevant in this case

**Rationale:** Most people are not carriers for rare disorders, the more rare the disorder is, the less likely a person is to be a carrier. PKU is a rare disorder. So, we know that the man has PKU, his genotype is (aa). His wife is not related to him (non-consanguineous), so her genotype is most likely to be AA. All of their children will most likely be carriers.

Later in this course (that is before exam #2) we will mathematically calculate that since PKU has an incidence of approximately 1 in 10,000 the probability of finding a carrier is 1 in 50. Therefor there is a 49 out of 50 chance that a person who appears healthy (phenotype), is a carrier with the Aa genotype for PKU. This is another level of math, and we will get to it in a few weeks... so hang on.

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**Question #: 9**

A scientist studying an autosomal recessive trait accesses data from publicly deposited genomic repositories to sample DNA sequences from over 250,000 individuals. A specific allele is found to have a frequency of approximately 1 in 100 in the general population. Based on these findings, which of the following is the best estimate of the frequency of homozygotes for this allele in the population?

- A. 1 in 250
- B. 1 in 5
- C. 1 in 50
- D. 1 in 500
- ✓E. 1 in 10,000
- F. 1 in 100
- G. 1 in 250,000

**Rationale:** The allele frequency is 1 in 100 (this is  $q$ ), so the frequency of homozygotes of that allele would be  $q$ -squared, which is 1 in 10,000. The frequency of finding a carrier would be 1 in 50 individuals. The vignette states that the sample size is 250,000 individuals, this is simply telling us that it is a relatively large sampling. A large set like this is not unreasonable given that some commercial companies such as 23andMe have many millions of individuals in their database.

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**Question #:** 10

A researcher studying Marfan syndrome finds that alleles of *FBN1* that are causative of the disorder are found at a frequency of approximately 1 in 10,000. Based on the data, which of the following best describes the incidence of the disorder?

- A. 1 in 2,500
- ✓B. 1 in 5,000
- C. 1 in 10,000
- D. 1 in 20,000
- E. 1 in 25

**Rationale:** Marfan syndrome is autosomal dominant, so the disease allele frequency will be 1/2 of the incidence of the disorder.

However, the allele frequency is given (1 in 10,000), so the number of people who have the disease allele in their genotype (and be affected) will be 2x the allele frequency.

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**Question #:** 11

An investigator is comparing the incidence of genetic disorders in the human population and selects only those disorders that have an allele frequency of 1 in 1,000. From these specific disorders which of the following would most likely be the rarest type of disorder expected in the population (meaning the least frequently observed)?

- A. A disorder caused by mitochondrial inheritance
- B. X-linked disorders
- C. Autosomal dominant disorders
- ✓D. Autosomal recessive disorders
- E. Multifactorial disorder

**Rationale:** For autosomal recessive, If the allele frequency is 1 in 1000 (q), then the incidence of the disorder would be 1 in a million ( $q^2$ ). For autosomal dominant, incidence would be 1 in 500; for X-linked about 1 in 1000 boys would be affected. If this hypothetical allele were inherited via the mitochondrial genome, 1 in a 1000 families would be affected.

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**Question #:** 12

A 20-year-old woman comes to the physician for genetic counseling because she is a known carrier of an allele associated with a rare autosomal recessive form of dystonia. Because it is rare, and challenging to diagnose, the disease incidence is difficult to estimate, however, it is thought to be approximately 1/250,000. What is the estimated risk that this woman will marry a carrier from the general population?

- A. 1/25
- B. 1/50
- C. 1/500
- ✓D. 1/250
- E. 1/5000

**Rationale:** For AR disorders the disease allele frequency is the square root of the incidence of the disorder, so q is approximately 1 in 500 for this disorder. The frequency of carriers in the population is twice this (all people have 2 autosomes each), so carrier frequency is 1 in 250.

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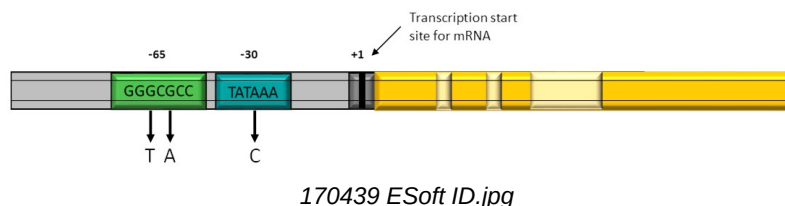
**Question #:** 13

A 3-year-old boy is brought to the physician for a follow-up examination after being diagnosed with beta-thalassemia. Genetic analysis shows that he has three different nucleotide substitution mutations within the promoter of the  $\beta$ -globin gene as shown by the vertical arrows in the figure. In vitro studies show that the mutations lead to reduced expression of the  $\beta$ -globin mRNA. Which of the following cellular events is most likely to be directly disrupted by these mutations?

- A. Coordination of ribosomes binding to mRNA
- B. Stimulation of poly(A) polymerase activity
- ✓C. Binding of basal transcription machinery to promoter
- D. Activation of histone acetylase activity
- E. Appropriate splicing of mRNA by snURPs

**Rationale:** This question could use any promoter in front of any gene. The promoter shown in the diagram has 3 point mutations in the promoter which can affect the ability of RNA polymerase and other elements of the basal transcriptional machinery to recognize and bind to the promoter region. In that case, RNA polymerase will not be as efficient as it could be and the result would be a reduction in the amount of mRNA being produced for this gene. This region of the promoter does not become part of the RNA (upstream of the transcriptional start site) and thus not effect ribosomal binding. The mutation is at the 5' end of the gene and thus would not be involved with the poly A polymerase activity which occurs at the 5' end of the gene (in the untranslated region). Histone and acetylase activity work at CpG islands and not part of/ near the protein encoding region. Again, the mutations are in the 5' untranslated region which is different from where snURPs bind in the regions that flank an intron.

**Attachment:**




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**Question #:** 14

A 58-year-old woman with a diagnosis of breast cancer comes to a follow-up appointment with oncologist. The physician asks the patient for consent to perform a biopsy of the tumorous tissue to analyze the gene expression profile in order to guide treatment decisions. The analysis will include the identification of microRNAs (miRNAs) which have a strong role in gene regulation. Which of the following best describes the function of miRNAs in gene expression?

- A. They encode proteins that directly regulate gene expression

- B. They bind to DNA to prevent transcription
- ✓C. They bind to target mRNA to inhibit translation
- D. They are involved in the splicing of pre-mRNA
- E. They act as enzymes to catalyze biochemical reactions

**Rationale:** MicroRNAs (miRNAs) are small, non-coding RNA molecules that regulate gene expression post-transcriptionally by binding to complementary sequences on target mRNAs. This binding can result in the degradation of the mRNA or inhibition of its translation into protein.

miRNAs do not encode proteins; they are non-coding RNAs.

miRNAs do not bind to DNA; they interact with mRNA.

miRNAs do not function as enzymes. miRNAs are not involved in the splicing of pre-mRNA; this is the role of small nuclear RNAs (snRNAs).

**Question #: 15**

A 75-year-old man comes to the emergency department because of a 2-week history of lower abdominal pain and severe dysuria. He is diagnosed with a severe urinary tract infection and is started on constant intravenous infusion of an antibiotic at a rate of 10 mg/min. The antibiotic has a half-life of 16 hours, a Vd of 12 L, and a clearance of 0.5 L/min. Which of the following will most likely be the plasma concentration of the drug at steady-state?

- A. 0.1 mg/L
- B. 0.2 mg/L
- ✓C. 20 mg/L
- D. 50 mg/L
- E. 80 mg/L

**Rationale:** Dosing rate = CL X C<sub>ss</sub>

Therefore:

$$C_{ss} = \frac{\text{Dosing rate}}{CL}$$

$$C_{ss} = \frac{10 \text{ mg/min}}{0.5 \text{ L/h}} = 20 \text{ mg/L}$$

**Question #: 16**

A 43-year-old woman is being treated for a urinary tract infection. For the past two days she has been receiving a continuous intravenous infusion of an antibiotic. The antibiotic follows first-order kinetics of elimination. The plasma concentration of the antibiotic has reached steady-state. On the third day of admission, a nurse misreads the physician's instructions and reduces the infusion rate by half. Which of the following will most likely be the consequence of this change in the infusion rate?

- A. The plasma antibiotic steady state concentration will increase 100%
- ✓B. The plasma antibiotic steady state concentration will decrease 50%
- C. The half-life of the antibiotic will decrease 50%
- D. The half-life of the antibiotic will increase 100%
- E. The clearance of the antibiotic will increase 100%
- F. The clearance of the antibiotic will decrease 50%

**Rationale:** Infusion rate = Clearance X C<sub>ss</sub>. Clearance is constant for drugs showing first-order kinetics. Therefore the steady-state plasma concentration (C<sub>ss</sub>), is directly proportional to the infusion rate. If infusion rate is halved, steady-state plasma concentration is halved.

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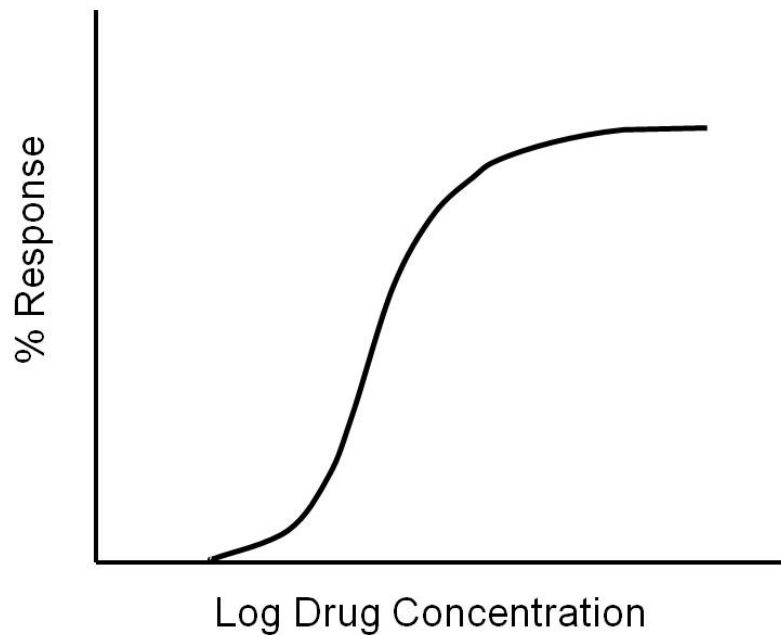
**Question #:** 17

A pharmacologist is studying the properties of a novel beta 2 agonist. He determines the degree of smooth muscle contraction evoked by the drug using an in vitro preparation of porcine airway tissue. The picture shows the dose response curve. Which of the following can most likely be obtained from this experimental curve?

- A. K<sub>d</sub>
- ✓B. EC<sub>50</sub>
- C. ED<sub>50</sub>
- D. Therapeutic index
- E. Therapeutic window
- F. Bioavailability

**Rationale:** The EC<sub>50</sub> is the concentration at which 50% of the maximal response of the drug is achieved. The EC<sub>50</sub> of a drug can be obtained from a graded concentration-response curve as shown in the figure.

**Attachment:**



BASICS 118.jpg

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**Question #:** 18

A 56-year-old woman is administered a 200-mg dose of an analgesic drug by intravenous bolus injection prior to a surgical procedure. The drug follows first-order kinetics of elimination with a volume of distribution of 120 L and a clearance of 14 L/h. Which of the following is the approximate percentage of the drug dose to be eliminated in 24 hours?

- A. 25
- B. 50
- C. 75
- D. 88
- ✓E. 94
- F. 97
- G. 99

**Rationale:** 50% of the drug is lost after 1 half-life, 75% after 2 half-lives, 87.5% after 3 half-lives and 93.75% after 4 half-lives. The half-life of the drug can be calculated as:

$$t_{1/2} = \frac{0.693 \times V_d}{CL} = \frac{0.693 \times 120 \text{ L}}{14 \text{ L/h}} = 6 \text{ h}$$

Therefore, in 24 h approximately 94% of the drug will be eliminated.

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**Question #:** 19

The pharmacodynamic properties of a novel antihypertensive drug, Drug X, is being studied in an in vitro preparation of rat renal blood vessels. The studies show that Drug X increases cAMP levels via activation of dopamine D1 receptors. Further studies show that drug X dose-dependently decreases the increase in cAMP levels evoked by dopamine, a full agonist at D1 receptors. Which of the following best describes Drug X?

- A. Physiologic antagonist
- B. Full agonist
- ✓C. Partial agonist
- D. Inverse agonist
- E. Non-competitive antagonist

**Rationale:** Drug X increases cAMP levels via activation of D1 dopamine receptors, therefore we can conclude that it is an agonist at D1 receptors. But Drug X decreases the effect evoked by dopamine, a full agonist at D1 receptors; therefore, Drug X must be a partial agonist. Recall that in the presence of a full agonist a partial agonist behaves as a competitive antagonist.

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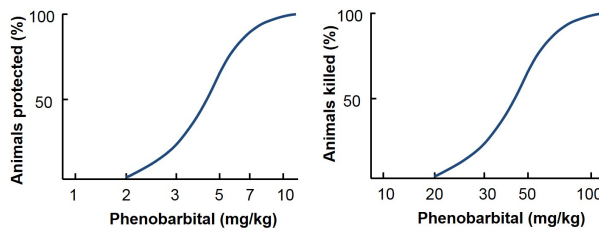
**Question #:** 20

A team of pharmacologists is determining the quantal dose-response curve for the protective effect of phenobarbital against convulsions in a group of rats. The figure shows the results of the experiment. Which of the following parameters of this drug is most likely obtained from such a dose-response curve?

- ✓A. The therapeutic index
- B. The bioavailability
- C. The EC50
- D. The Kd
- E. The Emax

**Rationale:** The therapeutic index is defined as the ratio of the TD50 or the LD50 to the ED50 for a therapeutically relevant effect. It is the dosage range between the minimum effective therapeutic concentration and the minimum toxic concentration.

**Attachment:**



BASICS TI.jpg

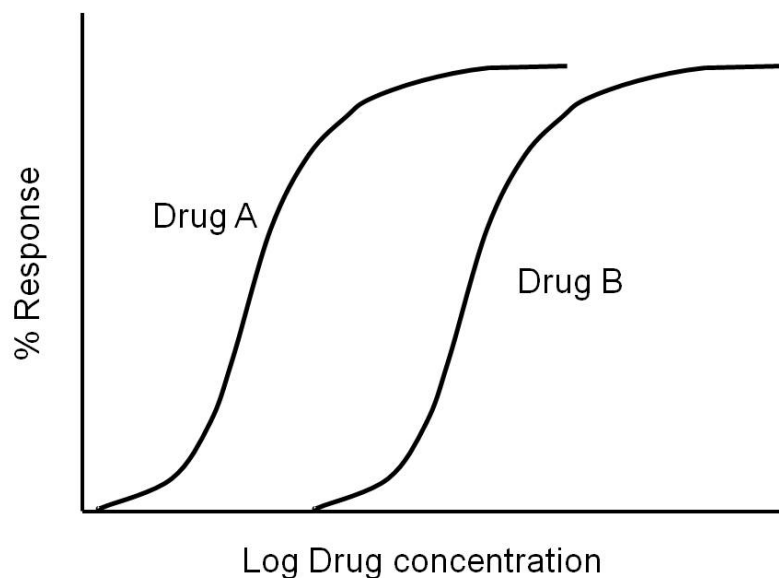
### Question #: 21

A pharmacologist is studying the properties of two novel drugs, Drug A and Drug B. Both drugs bind selectively and with high affinity to beta-1 adrenergic receptors. The graph shows the effect of drugs A and B on the contractile force of an in vitro preparation of guinea pig cardiac smooth muscle. Which of the following best describes the properties of these drugs?

- A. Drug A has the same potency as Drug B
- ✓B. Drug A is more potent than Drug B
- C. Drug B is more potent than Drug A
- D. Drug A is less efficacious than Drug B
- E. Drug A is more efficacious than Drug B
- F. Drug A is a partial agonist

**Rationale:** Potency is a measure of the concentration or amount of drug necessary to produce an effect of a given magnitude. The graph shows that Drug A can produce 50% of the maximal response at a lower concentration than Drug B, therefore it can be concluded that Drug A is more potent than Drug B.

**Attachment:**

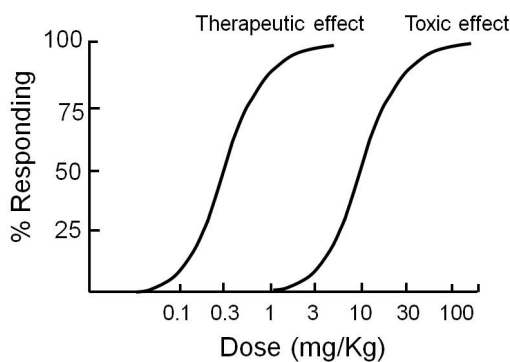


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**Question #: 22**

The effects of a novel drug were tested on a population of mice with experimentally-induced cardiac failure. The therapeutic effect was a 15% decrease in blood volume. The toxic effect was a 30% increase in heart rate. From the results obtained, the ED<sub>50</sub> and TD<sub>50</sub> were determined. The figure shows the quantal dose-response curves. Which of the following statements best describes the results of the study?

- ✓A. The therapeutic index is 33
- B. The therapeutic index is 10
- C. The ED<sub>50</sub> is 1 mg/kg
- D. The TD<sub>50</sub> is 30 mg/kg
- E. The therapeutic window is 1 - 30 mg/Kg

**Attachment:**

BASICS 92.jpg

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**Question #: 23**

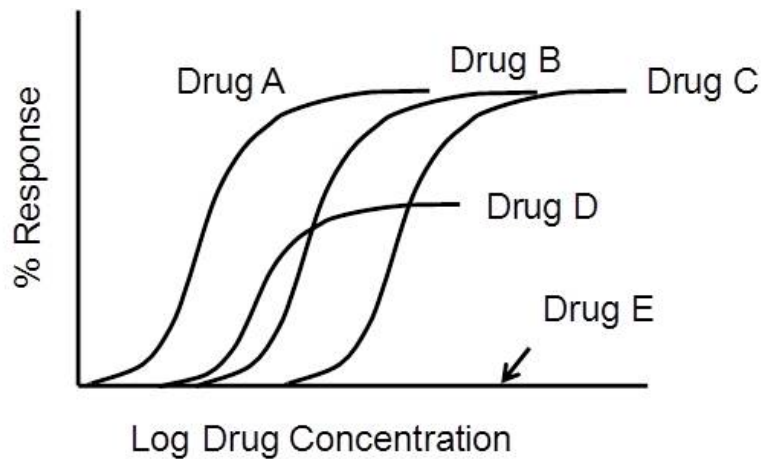
The effects of five new drugs on gastrointestinal smooth muscle contraction are being studied. The five drugs show high affinity for the M3 muscarinic receptor. The figure shows the experimental results. Which of the following drugs is most likely an antagonist?

- A. Drug A
- B. Drug B
- C. Drug C

- D. Drug D
- ✓E. Drug E

**Rationale:** An antagonist is a drug that blocks or reduces the action of an agonist.

**Attachment:**



BASICS 84.jpg

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**Question #:** 24

A team of pharmacologists is studying the pharmacokinetic properties of a novel antihypertensive drug. They desire to estimate the bioavailability of the drug after oral administration. Which of the following is most likely needed to determine their desire?

- ✓A. Area under the plasma concentration-time curve (AUC)
- B. Peak plasma concentration
- C. Half-life
- D. Time to peak plasma concentration
- E. Volume of distribution

**Rationale:** Bioavailability is the fraction of the administered dose of a drug that reaches the systemic circulation. By plotting plasma concentrations of the drug vs. time, the area under the curve (AUC) can be measured.

The AUC reflects the extent of absorption of the drug. The oral bioavailability is determined comparing the AUC after oral administration with the AUC after intravenous injection.

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**Question #:** 25

A 45-year-old man is brought to the emergency department by ambulance 2 hours after ingesting an overdose of an opioid analgesic. His blood pressure is 102/68

mm Hg, pulse is 56/min, and respirations are 11/min. Physical examination shows a semicomatose man with cool, damp skin and pinpoint pupils. Laboratory studies show that the plasma concentration of the drug is 2 mg/L. The drug follows first-order kinetics of elimination with a half-life of 3 hours. Which of the following is most likely to be the time required for the plasma drug concentration to reach a safe concentration of 0.5 mg/L?

- A. 3 hours
- B. 4 hours
- ✓C. 6 hours
- D. 9 hours
- E. 10 hours

**Rationale:** Half of the dose of the drug is eliminated every half life. The half-life of the drug is 3 hours and the current plasma concentration is 2 mg/L. Therefore, after 1 half-life the concentration will be 1 mg/L and after 2 half-lives it will be 0.5 mg/L.

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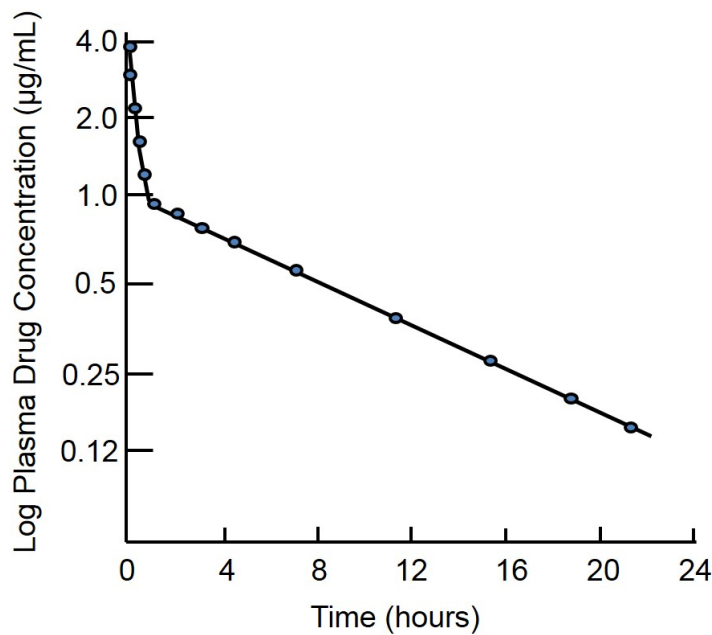
**Question #:** 26

A 200 mg dose of a new drug was administered as an intravenous bolus to 50 healthy volunteers, and blood samples were analyzed at intervals. The graph shows the average plasma concentration over time. Which of the following values is the best approximation of the calculated volume of distribution of the drug?

- A. 0.5 L
- B. 1 L
- C. 50 L
- D. 100 L
- ✓E. 200 L

**Rationale:** This question requires students to be able to interpret the graph given and by extrapolation estimate the concentration of the drug at time 0 if it is immediately distributed. We can extrapolate and find that the concentration of the drug at time 0 is 1 microgram/ml which is equal to 1 mg/L. We can plug this number to the formula of Vd ( $V_d = \text{Dose} / \text{Concentration}$ ) which will give us a  $V_d = 200\text{L}$

**Attachment:**



BASICS 82.jpg

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**Question #: 27**

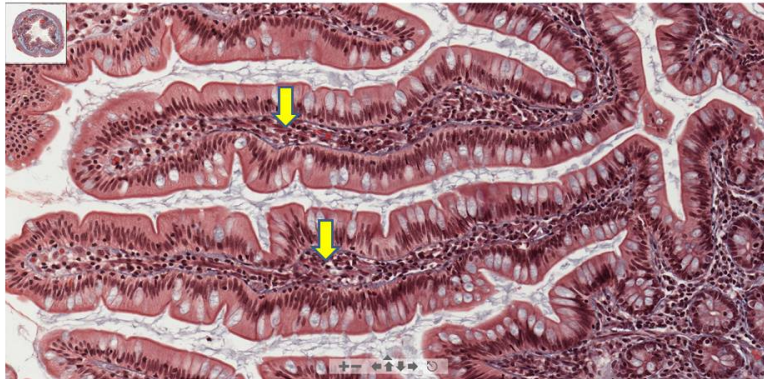
A 28-year-old woman is brought to the physician because of an 8-day history of abdominal pain, and weakness, and a 3-month history of intermittent nausea, vomiting, and weight loss of 6.8 kgs (15 lbs). Physical examination shows an emaciated woman with dried oral mucosa and skin. An endoscopy shows inflammation of the small intestine. A biopsy is shown in the attached image. Which of the following best describes the function of the connective tissue indicated at the tip of the arrows?

- A. Imparts elasticity to the tissue
- B. Withstands unidirectional stress
- C. Serves as energy reserve
- ✓D. Protection from invading organisms
- E. Allows large volume changes in organs

**Rationale:** The arrows in the image are pointing to the highly cellular and vascularized tissue (loose connective tissue), directly beneath the epithelium of the intestinal mucosa. Loose connective tissue is primarily located beneath the epithelia that cover body surfaces and line internal surfaces. This tissue is thus the initial site where pathogenic agents, such as bacteria that have breached an epithelial surface, are challenged and destroyed by cells of the immune system (Option D), which are mostly transient wandering cells that migrate from local blood vessels in response to specific stimuli. Loose connective tissue, therefore, is the site of inflammatory and immune reactions. For example, the lamina propria, the loose connective tissue of mucous membranes, such as those of the respiratory and alimentary systems, contains large numbers of these cells.

- Option A- elastic tissue.  
Option B- Dense Regular CT.  
Option C- adipose tissue.  
Option E- Dense Irregular CT.

**Attachment:**



Picture12.JPG

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**Question #:** 28

A 52-year-old woman comes to the physician because of a 3-month history of pain in her lower back and hips, and unintentional weight loss of 6.3 kgs (14 lbs). **Her vital signs are within normal limits.** Physical examination shows tenderness over her hips. Laboratory studies show hypercalcemia (elevated serum calcium levels) and thrombocytopenia (low platelet count). A bone marrow biopsy shows over proliferation of the cell shown in the attached images. Which of the following best characterizes a function of this cell?

- A. Secretion of connective tissue matrix
- ✓B. Production of antibodies
- C. Secretion of elastic fibers
- D. Production of reticular fibers
- E. Synthesis of collagen fibers
- F. Thermogenesis

**Rationale:** The images shown represent plasma cells which are derivatives of B lymphocytes of the wandering/transient cell population of connective tissue. They are recognized by their characteristic nucleus of peripheral heterochromatin with alternating euchromatic regions ('clockface appearance') as

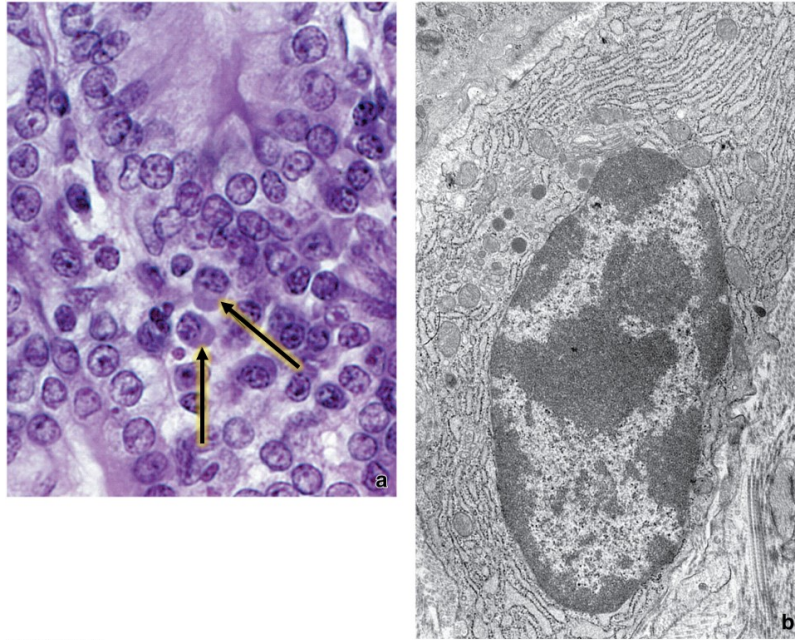
well as prominent Golgi apparatus (negatively stained in light micrograph).

Plasma cells produce antibodies (proteins) and therefore also have extensive, Rough ER (seen in TEM).

Fibroblasts secrete CT matrix and fibers.

Adipocytes are responsible for thermogenesis

**Attachment:**



PlasmaCell.jpg

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**Question #:** 29

A 33-year-old woman comes to the emergency department because of a 48-hour history of absent fetal movements. She is at 37-weeks gestation. Physical examination shows absent fetal heart rate. She subsequently delivers a stillbirth fetus. A biopsy of the umbilical cord shows normal blood vessels surrounded by connective tissue. Which of the following numbered images best represents the connective tissue type in the biopsied structure?

- A. 1
- B. 2
- C. 3
- D. 4
- ✓E. 5
- F. 6

**Rationale:** The CT in the umbilical cord is Mucous CT (Option 5), characterized by few cells and thin, wispy collagen fibers, and abundant ground substance that is gelatinous due to high content of hyaluronic acid, (prevents kinking of the blood vessels in the umbilical cord - allows for adequate materno-fetal blood flow).

Option 1- Dense Irregular CT (more fibers than cells, fibers haphazardly arranged).

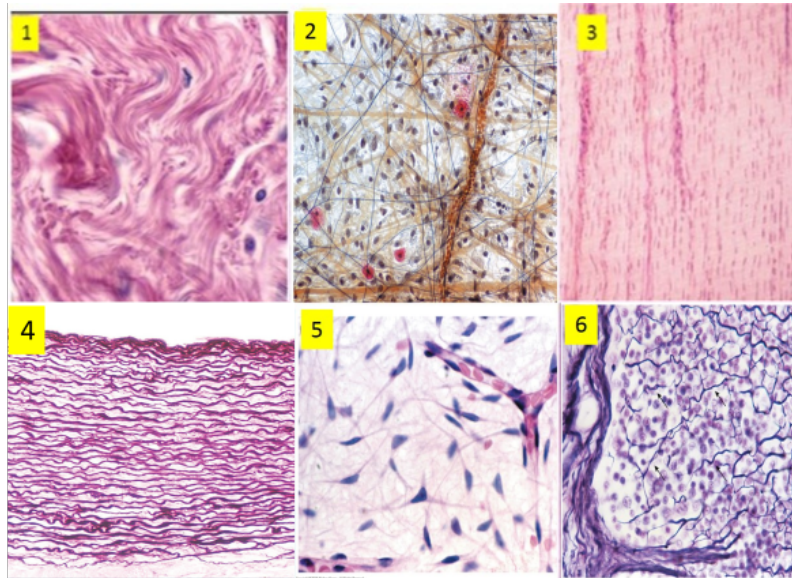
Option 2- Loose CT (more cells than fibers).

Option 3- Dense Regular CT (more fibers than cells, with fibers densely packed in parallel arrays to resist unidirectional stress).

Option 4- Elastic fibers- dark red wavy lamellae(highlighted with a special elastic stain) in walls of an elastic artery.

Option 6- Reticular tissue/fibers (argyrophilic i.e silver-loving; short, thin and extensively branched fibers).

**Attachment:**



Connective Tissue collage1red1.png

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**Question #:** 30

A 43-year-old man comes to the physician because of a 1-month history of a swelling on his right wrist. Physical examination shows a firm, non-tender mass around the tendon of the flexor carpi ulnaris (forearm muscle). A biopsy shows a mass with infiltration into the dense connective tissue of the skin. Which of the following images best features the histological appearance of the infiltrated tissue?

- ✓A. 1
- B. 2
- C. 3
- D. 4
- E. 5
- F. 6

**Rationale:** The infiltrated tissue is the dense connective tissue of the skin (which is the deeper 'reticular layer'). This dense CT in skin needs to be able to resisting shearing forces from all directions and is therefore characterized by having more fibers than cells, with ahaphazard arrangement of fibers) (Option

1- Dense Irregular CT).

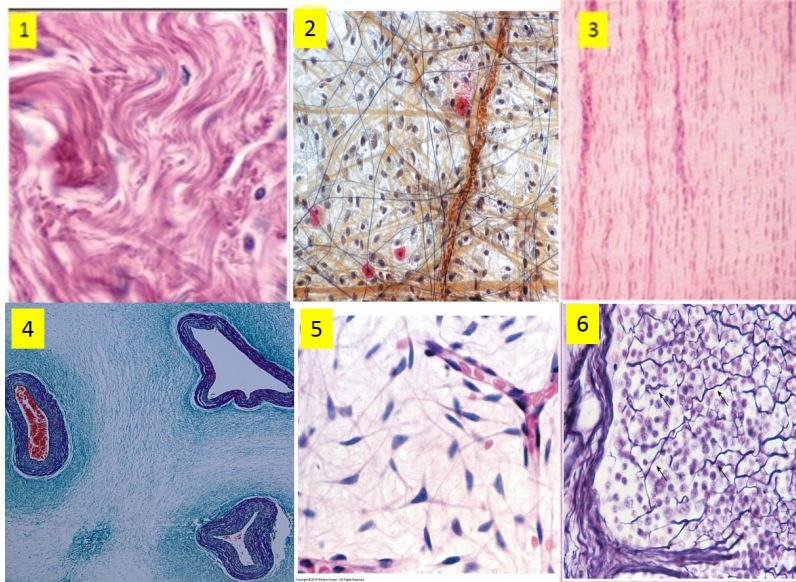
Option 2- Loose CT (more cells than fibers).

Option 3- Dense Regular CT (more fibers than cells, with fibers densely packed in parallel arrays to resist unidirectional stress).

Options 4 & 5- Embryonic CT (Abundant ground substance, scanty fibers).

Option 6- Reticular tissue/fibers (argyrophilic i.e silver-loving; short, thin and extensively branched fibers)

**Attachment:**



*connective tissue collage .jpg*